UNIVERSITY OF BIRMINGHAM DUBAI

برنامج الذكاء الاصطناعي
THE ARTIFICIAL INTELLIGENCE PROGRAM

LEARN AI
STUDENT AGENT

The Personalized Student Agent

What information (data) does an AI agent need to monitor students' academic progress and develop personalized learning plans that address their individual learning needs?

The Mission

You are developing an AI agent to monitor academic progress and create personalized learning plans for 10,000 students aged 13–16 across 20 schools in the UAE.

Your agent is responsible for:
- Reviewing students' performance across assignments, examinations, quizzes, and learning activities.
- Identifying areas of strength, knowledge gaps, and recurring misconceptions.
- Developing personalized learning plans based on each student's academic progress and needs.
- Recommending and generating tailored learning materials, exercises, and assessments.
- Monitoring how students respond to recommended content and adjusting their learning plans accordingly.



Section 1: What does the agent need to know?

This section is focused on thinking about what information your agent needs to operate on. For now, do not think about *preprocessing*, but just what *data sources* your agent needs.
For each data item, mention:
- **where** this information will come from
- **how** often it must be updated
- **why** the agent needs this data

*Think about what data your agent needs from **each student's academic performance and progress**. What d will the agent need from assignments, examinations, quizzes, and previous learning activities to identify areas where the student needs additional support? How would you **collect it**?*

*Think about what data your agent needs from the **curriculum and available learning resources**. What information will the agent need to decide which topics to prioritize, what content to recommend or generate, and how to structure an appropriate learning plan? How would you **collect it**?*

*Think about what data your agent needs from **students' interactions with learning materials and activities**. What information will help the agent understand whether its recommended content is effective and whether the student's understanding is improving? How would you **collect it**?*

Section 2: From Raw Data to Actionable Information (Data Preprocessing and Representation)

Your agent now has access to the data sources you identified in Section 1 related to students' academic progress, curriculum and learning resources, and learning interactions. However, raw data is not necessarily ready for an AI agent to use. Some observations may be incomplete, noisy, duplicated, outdated, recorded in different formats or grading systems, or too large to process efficiently.

Design a preprocessing pipeline that transforms your collected data into reliable, compact, and useful information for the agent's decisions.

Consider the data sources you identified in Section 1. For three different data sources, answer the following questions:

1. What **data-quality problems** might you encounter? Consider missing assignment submissions, inconsistent examination grading systems, duplicate records, incorrect answer keys, and outliers.
2. What **preprocessing techniques** would you apply to address these problems? Be specific: would you standardize grades across different assessments, remove duplicate records, correct inconsistent labels, or flag suspicious observations?
3. When might preprocessing introduce **errors**? ***Think carefully***: Should missing assignment grades be filled in? Could standardizing examination scores hide differences in difficulty? When is it better to preserve missing information and communicate uncertainty about a student's understanding to the agent?

[Data Source 1] _____

[Data Source 2] _____

[Data Source 3] _____

If different sources need to be combined, how would you **align them**? For example, matching student identifiers across school systems, linking assignment and examination questions to the same curriculum concepts, or connecting recommended learning activities with subsequent assessment results.

Some of your chosen sources may contain **large amounts of information**, such as textbooks, recorded lessons, handwritten assignments, examination papers, or extensive histories of student interactions.

The agent cannot necessarily process or store all this information in its original form. Identify one or two data sources that would be particularly expensive to store, transmit, or process.

What information from these sources would actually be useful to the agent?

How would you transform the raw data into a smaller, structured representation? For example, could examination papers and assignments be converted into question-level performance records linked to specific curriculum concepts? Could textbooks and learning materials be transformed into organized topics, prerequisite relationships, difficulty levels, and searchable learning activities?

What information might be lost through compression, aggregation, or feature extraction? Under what circumstances would you need to retain or revisit the original data?

Section 3: Memory and Data During Deployment

Your personalized student learning agent is now operational across 20 schools in the UAE. Every day, students complete assignments, take examinations, interact with recommended learning materials, and receive new learning plans. These activities generate new information about students' academic progress, the effectiveness of recommended content, and their changing learning needs.

Designing your Agent's Memory

What information should your agent remember about individual students, their academic progress, and previous learning plans to make better recommendations in the future?

Design **three types of memory**. Specify **what information each contains**, why it would help the agent **make better decisions over time**, how it is **updated**, and when observations should expire or be revalidated.

[Memory 1] _____

[Memory 2] _____

[Memory 3] _____